

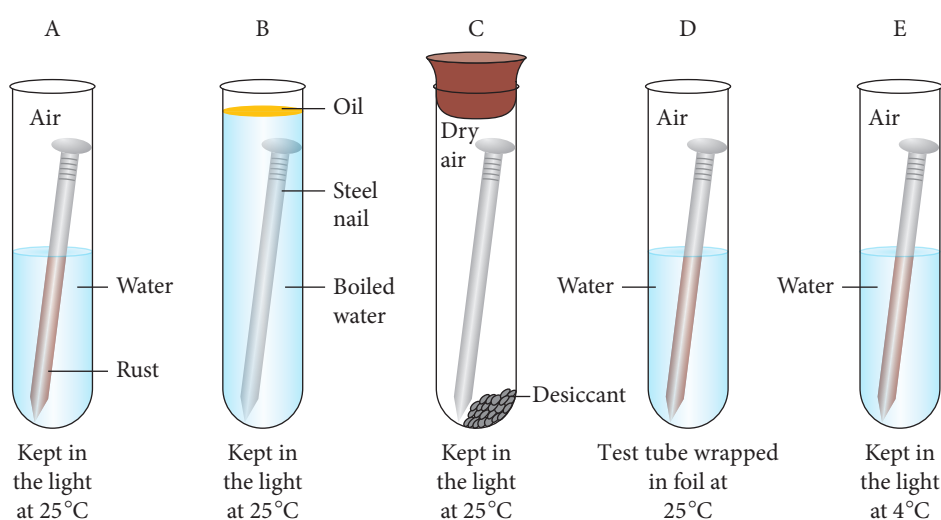
Name: _____

Class: _____

ACTIVITY SHEET

4.5 Analysing experimental designs

Steel nails left outside tend to rapidly rust unless they have been galvanised. A student designed an experiment to identify the four factors (air, moisture, light and warmth) that are required for steel nails to rust. Five test tubes were set up and the results after 48 hours are shown below.



- 1 What was the purpose of test tube B?
- 2 How did the experimenter ensure that no air reached the nail in test tube B?
- 3 What component of air might be involved in the rusting process?
- 4 What was the purpose of test tube C?
- 5 How did the experimenter ensure that no moisture reached the nail in test tube C?

- 6 What was the purpose of test tube D?
- 7 How did the experimenter ensure that no light reached the nail in test tube D?
- 8 What was the purpose of test tube E?
- 9 How might the experimenter have ensured the nail in test tube E was kept illuminated at 4°C?
- 10 What was the purpose of test tube A?
- 11 Would test tube A qualify as a control?
- 12 What does test tube A by itself show?
- 13 What does test tube A by itself not prove?
- 14 What conditions are definitely required for rusting to occur?
- 15 How do you know that?
- 16 Which condition is definitely not required for rusting to occur?

17 How do you know that for sure?

18 Can you assume from the experiment that rusting would occur at very low temperatures? Why?

19 How could you investigate that?

20 How could the experiment be improved?